S.P.I.T. — Early Wildfire Prevention System

Climate change is amplifying fire-conducive conditions, increasing the likelihood of ***extreme fire behavior*** in regions that historically were less vulnerable, including boreal and temperate ecosystems. NASA highlights that extreme wildfires have become more frequent, more intense, and larger in recent decades, notably across temperate conifer forests and boreal zones. At the same time, global burned area can decline while societal impact rises: a 2025 *Science* study reports a **26% decline in global burned** area (2002–2021) yet a ~40% increase in human exposure, driven by the growing co-location of fires and settlements. **Spain's 2025** season illustrates this escalation in impact, with **~380,000 ha burned**.

To reduce this gap between *burned area* and *consequences*, we developed S.P.I.T. (Early Wildfire Prevention System) for high-risk forest interfaces: a connected, rapid-response architecture that **combines continuous detection with remote actuation** of a *wet fuelbreak* to confine ignition before it transitions to a fast-spreading front. The system is unified under 5G communications and a centralized command post at the fire station, where alerts are received, validated, and converted into an immediate suppression action.

Phase 1 — Prevention & Early Warning. A UAV (24/7 surveillance) fleet patrols in shifts to maintain uninterrupted coverage of high-probability ignition zones. Each drone integrates RGB + thermal imaging and runs on-board inference TinyML deployment on the flight controller, **transmitting events** instead of continuous video. This design reduces uplink bandwidth, lowers end-to-end latency, and improves **operational scalability** when multiple sectors are monitored simultaneously. When a thermal anomaly consistent with incipient combustion is detected, the drone sends an alert to the command post for confirmation and response coordination.

Phase 2 — Remote wet fuelbreak for confinement. Once active fire is confirmed, SPIT activates a sprinkler network installed along existing firebreak corridors, wetting the lateral boundaries of the affected sector with EcoFire, selected as a **lower-environmental-impact alternative** to PFAS-based foams, which are widely documented sources of persistent PFAS contamination in soil and water. EcoFire is specified as an aqueous solution of organic loads intended to act as a fire-extinguishing and fire-barrier agent.

Why the **barrier works**: Fire propagation requires sustained heat transfer to unburned fuel and continuous pyrolysis. A wet fuelbreak suppresses spread by (1) **increasing effective heat capacity** and evaporation demand at the boundary, (2) improving penetration into porous fuels, and (3) **reducing fuel continuity** and delaying ignition, effectively disrupting the fire at the perimeter. EcoFire's barrier claim aligns with this mechanism acting as a barrier against fire. Hydraulically, EcoFire is denser than water: 1.145 kg/m³ and Newtonian viscosity 27.40 ± 0.04 mPa·s at 25 °C. These properties increase pressure losses versus water, so the design uses **progressive cavity pumps** to maintain stable flow with viscous liquids and dosing control across elevation changes, controlled by a **PLC with pressure & flow feedback.**

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For a 10 ha critical area, the deployment includes ~400 sprinklers spaced ~3.2 m, a 10,000 L EcoFire buffer sized for a **rapid 30-minute** perimeter wetting event. Project value for 10 ha corresponds to €43k. The intended outcome is to convert a high-growth incident into a confined, lower-intensity event, buying the minutes that dominate fire escalation dynamics and **improving firefighter safety and biodiversity protection**.

Datasets

Pump Design

Parameter	Value	Unit
Required sprinkler pressure	3	bar
Sprinkler head	30.591	mH_2O
Total friction loss	31.041	mH_2O
Filter/valve losses	5	m
Elevation difference (static lift)	100	m
Total required head (H)	71.630	m
Equivalent pump pressure	7.025	bar
Design flow rate (Q)	105.410	L/min
Flow rate (Q)	6.325	m^3/h

Hydraulic System per 10 ha

Parameter	Value	Unit
Volume V per activation	3162.30	L
Time t	30.00	min
Average total flow Q_total	105.41	L/min
No. of simultaneous sectors	1.00	units
Total sprinklers	400	units
Active sprinklers per sector	100	units
Required flow per sprinkler	1.0541	L/min

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